Welcome to your next lesson: Composition vs inheritance: a practical decision rule.

I'm preparing your personalized lesson based on what you've already learned, what's next, and how you learn best.

Step by step, you’re moving forward.

Imagine two things in the pets world:

A PetOwner and a Pet
A Dog and a Pet 🐕
Think in plain English:

A PetOwner has a pet.
A Dog is a pet.

So:

"has-a" → one object contains/uses another as a part (like an owner using a pet).
"is-a" → one thing is a special kind of another (Dog is a kind of Pet).
Quick gut check for you, Ron:

Which sentence sounds right for PetOwner and Pet?

"A PetOwner has a Pet"
or
"A PetOwner is a Pet"

You can just answer with the sentence you think fits better.

has a pet

What do you think? Pick the best option in the question below 👇

A PetOwner stores and uses a Pet object. Which relationship is this?

A.
Has-a/composition


B.
Is-a/inheritance


C.
Polymorphism

Exactly right 🎯

A PetOwner that stores and uses a Pet is a has-a relationship, also called composition.


The owner is not a kind of pet.
Instead, the owner has a pet as part of their data and behavior (maybe feeds it, walks it, etc.).

So in design terms: PetOwner has-a Pet → use composition.

Next small step 👣


Now let’s flip it:


Think about Dog and Pet.


In plain English: does


"A Dog is a Pet"
sound right, or
"A Dog has a Pet"?

Reply with whichever sentence you think fits better.

is a pet

What do you think? 🕵️ Pick the best answer in the question above 👆

The Pet Design Detective finds that a Dog is a specialized kind of Pet. Which design choice earns the badge?

A.
Composition

B.
Inheritance

C.
Neither relationship

Nice catch, Ron — that’s exactly it 💪

If a Dog is a Pet, that means:

Dog is a specialized kind of Pet 🐕
So in code, Dog should inherit from Pet:
Python

```
class Pet:
    pass

class Dog(Pet):  # Dog is-a Pet
    pass
```

So your choice Inheritance is spot on ✅

Before we wrap this lesson:

Do you want a super quick, concrete recap of the rule (“when to use has-a vs is-a”), or are you feeling ready to move on to the next lesson?

I'm ready to move on

Love it 🚀 Moving you forward.

Quick recap in plain language:

Has-a / composition → one object contains or uses another.

Example: PetOwner has a Pet → PetOwner stores a Pet object inside.
Is-a / inheritance → one class is a specialized version of another.

Example: Dog is a Pet → class Dog(Pet): ...

Simple test you can use anytime:


If the sentence “A is a B” feels right → consider inheritance.
If “A has/uses a B” feels right → choose composition.

You nailed those choices today 👏


When you’re ready, hit Next Lesson in the Learn tab and we’ll keep building on this.

How would you rate this lesson?